

Feasibility Memo: Yale EMLLC Triage Extract for Mercer General ED

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1. Verdict

Proceed with caveats: the dataset is suitable for developing a baseline model, while recognising that the dataset should first be validated against Mercer General's patient populations to ensure it is representative

2. Dataset Summary

The extract holds 55,121 adult emergency department (ED) encounters and 226 columns. One column is simply a row of numbers added when the data was exported and does not contain meaningful information. Around 200 columns indicate different chief complaints and the remaining columns contain patient identifiers, vital signs and other administrative information.

Metric	Value
Encounters	55,121 ED arrivals
Columns	226 raw data. 200 are chief complaints
Age Range	18-107 years (mean 55.3 , std 19.5)
Gender	Female = 31744 (57.6%) , Male = 23377 (42.4%)
Missing Emergency Severity Index (ESI) Labels	0 every encounter has a triage level assigned
Missing Data Overall	0 of 26 structured columns have any gaps
Disposition	Discharge = 34565 (62.7%) , Admit = 20556 (37.3%)

The distribution of ESI categories is highest among the middle severity levels, which is consistent with the patient mix typically seen in a general ED:

ESI Level	Encounters
1. Resuscitation	77 (0.1%)
2. Emergent	17,924 (32.5%)
3. Urgent	27,010 (49.0%)
4. Less urgent	8,896 (16.1%)
5. Non-urgent	1,214 (2.2%)

After cleaning the data, the patients' initial vital signs were generally within the expected clinical ranges. For example, the average heart rate was 86.4 bpm, mean systolic blood pressure 133.7 mmHg, mean respiratory rate 17.8/min and mean temperature 98.1 °F (36.7 °C).

3. Top 3 Concerns

- **Unusually complete data:** None of the 26 main data fields contained missing values, which is uncommon for real ED records. This means that the data was cleaned or missing values were filled in before the dataset was released. Mercer General's own data should not be expected to be this complete, however, there is a process step that handles missing values such as filling in typical values and creates "Unknown" categories so that the system can continue working with less complete real-world data.
- **Avoiding the outcome information influencing predictions:** The disposition and previous disposition field describes events that occur after the initial triage process. Because these details are only known later in the patient's visit, they were excluded from all data reviews and cleaning steps to prevent the model from using future information to make predictions.
- **How representative the data is:** The dataset comes from a US hospital and may not reflect the patient population and Mercer General. The datasets race, ethnicity and insurance pattern reflect the US healthcare system and it has not been compared with Mercer General's own data.

4. Top 3 Reasons to Proceed

- **A complete and reliable triage label:** All 55,121 ED visits have an assigned ESI level during clinical assessment. No labels were missing or needed to be estimated. This ensures that the model is learning from real triage decisions rather than using indirect substitutes.
- **Vital signs and patient complaints follow expected severity patterns:** Oxygen saturation is the vital sign strongly linked with ESI level. The r value of 0.178 means there is a small positive relationship between the two. Chest pain ($r = -0.164$) and shortness of breath ($r = -0.150$) are the strongest complaint-level indicators of higher acuity, followed by suicidal concerns (-0.143), alcohol intoxication (-0.142) and altered mental status (-0.132). These are all complaints that triage systems typically treat as urgent.
- **Clear and repeatable data preparation process:** All data cleaning decisions are documented in a clean triage process along with the reason behind each step. This allows others to review and reproduce the process rather than relying on unexplained decisions.

5. Caveats

- **Temperature units:** Temperature is recorded in Fahrenheit (°F), not Celsius (°C) and both versions are kept separate to avoid mixing scales.
- **Rare Complaint indicators:** Most of the 200 complaint columns appear in very few visits and add little value on their own, so the rarest values will be removed or grouped before modelling.
- **Correlation does not equal prediction:** The links between vital signs, complaints and ESI levels show patterns in the data, but they do not confirm how accurately these factors can predict a patient's triage level.

- **Sensitive complaint categories:** Some strong predictors of higher acuity involve mental health and behavioural complaints. These reflect clinical urgency and not patient judgement.
- **Applicability to a Caribbean ED:** This model has not been tested in a Caribbean setting and the US dataset may differ from Mercer General's patient population. A comparison should be done before using it in practice.

Top 10 Features Shortlist

The following features were ranked by their correlation with ESI from data profiling analysis. Information that would be available after triage such as the disposition and previous disposition was excluded. This list is a starting point for further testing not a ranking of the most important predictors as correlation only shows a relationship but does not show the importance.

Rank	Feature	Relationship with ESI
1	Oxygen Saturation	+ 0.178
2	Chest Pain Complaint	- 0.164
3	Shortness of Breath Complaint	- 0.150
4	Suicidal Thoughts Complaint	- 0.143
5	Alcohol intoxication Complaint	- 0.142
6	Altered Mental Status Complaint	- 0.132
7	Psychiatric Evaluation Complaint	- 0.103
8	Respiratory Rate	- 0.095
9	Heart Rate	- 0.095
10	Weakness Complaint	- 0.081

A negative correlation means that as the feature is more common, the ESI value tends to decrease, which corresponds to higher urgency levels because lower ESI levels represent more severe cases. Oxygen saturation works differently: a positive correlation is expected because lower oxygen levels indicate greater clinical concern and lower oxygen saturation values are associated with urgent cases.